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Syntactic Comprehension—A Separate Source of Individual Variance in Middle-School Children's Reading Comprehension

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ABSTRACT

The present study examined whether syntactic comprehension constitutes a source of individual differences separate from vocabulary and of relevance to reading comprehension. One hundred and sixty-one Danish Grade 6 students completed multiple tests of syntactic comprehension, vocabulary, decoding, and reading comprehension. Syntax measures were designed to isolate syntactic skills. Syntax and vocabulary measures included expressive and receptive tasks to balance method variance. Structural equation models supported that syntactic comprehension and vocabulary constituted separate, moderately correlated factors, and that syntactic comprehension had the strongest direct effect on reading comprehension. The results suggest that syntactic comprehension and vocabulary are separate sources of individual differences, and thus that some students misunderstand texts due to difficulties establishing how individual words relate to each other in sentences based on syntactic signals. The study also suggests that for sixth-grade students, individual differences in syntactic comprehension are at least as important as individual differences in vocabulary.

1 | Introduction

Despite disagreement about the details across diverse linguistic theories, syntax is recognized as a central part of how languages convey complex meaning in spoken or written language (Chomsky 2002; Dik and Hengeveld 1997; Harder 1997; Matthews 1981), for example, by establishing that "John" did the seeing in the passive sentence "Peter was seen by John." A vast number of studies in psycho- and neurolinguistics show that people at least partially process the syntactic structure of sentences when comprehending them (see Macdonald and Hsiao 2018; Osterhout et al. 2012 for overviews). Theories of reading comprehension recognize a role for syntactic processes in reading comprehension (Duke and Cartwright 2021; Perfetti and Stafura 2014), but relatively little attention has been paid to this role. For example, it is unclear if inaccurate use of syntactic information is a bottleneck in a substantial number of children's reading comprehension. Most children may have developed sufficient syntactic skills by school age to allow good comprehension. However, an increasing number of recent studies have found correlations between individual differences in syntactic comprehension and reading comprehension after plausible controls, suggesting that there is practically relevant variation in syntactic skills (e.g., Authors 2022; Silva and Cain 2015; Sorenson Duncan et al. 2021; Tong et al. 2024). These studies have, however, been limited by using single indicators of syntax and control variables, and so the relationship between syntax and reading comprehension has not been clearly established. The present study examined the construct of syntactic comprehension relative to vocabulary, decoding, and reading

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comprehension by employing latent variable modeling with multiple measures of each potential construct.

1.1 | The Role of Syntax in Comprehension

Syntax can be considered a subcomponent of grammar alongside morphology. Lexical morphology concerns building word meanings through compounding (*earth-quake*), inflection (*car-s*), and derivation (*walk-able*). Syntax—on the other hand—concerns combining lexical meanings into larger phrases and ultimately propositions. For example, in the sentence "The old woman saw the man with the beautiful flowers," it is not enough to know the meaning of "old," "woman," "saw," "man," "beautiful," and "flowers." One must also be able to establish the meaning relations between words based on syntactic signals such as word order. In the example above, word order not only signals who is doing what to whom ("the woman" is the one seeing, and "the man" is the one being seen), it also signals how words combine into meaning units ("the man" has "the beautiful flowers," "the woman" is "old," and "the flowers" are "beautiful"). In the same way, syntactic signals can also indicate what happens when, where, and why. In the sentence "the woman, who I met this morning, saw the man as she crossed the parking lot on her way to the store," word order signals what happened "this morning" (the speaker met the woman), and what happened as the woman crossed the parking lot (she saw the man). Sometimes, such relations may be inferable from context and background knowledge, but even then, robust understanding of the syntactic signals will help consolidate the literal meaning of the sentence. This implicit knowledge of how syntax expresses meaning relations between words has been denoted by various terms such as *syntactic knowledge*, *sentence comprehension*, or simply *syntax*. In this paper we use the term s*yntactic comprehension* in line with MacKay et al. (2021).

As shown above, syntactic comprehension is a principal part of understanding the propositional content of sentences together with the knowledge of individual words (i.e., vocabulary). However, it is less clear if some children experience reading comprehension difficulties because they have difficulties using syntactic signals to establish how the individual words relate to each other: The example sentences above may not pose comprehension difficulties for the vast majority of typically developing school-aged children. However, in school texts, especially from middle school and onward, texts grow increasingly syntactically complex, for example, containing longer nominal phrases and more subordination (Fang 2006; Uccelli et al. 2015). Furthermore, children will sometimes only have little background knowledge of these academic subjects, making it more difficult to guess or infer how word meanings relate to each other in complex sentence meanings. Thus, successful reading comprehension may become increasingly dependent on the ability to make use of syntactic signals such as word order to help combine the meaning of individual words into meaningful propositions.

And indeed, the majority of recent studies find that syntactic comprehension uniquely predicts variance in reading comprehension after controlling for vocabulary and additional controls, ranging from 5% to 12% (Gottardo et al. 2018; Poulsen and Gravgaard 2016; Authors 2022; Silva and Cain 2015; Sorenson Duncan et al. 2021; Tong and McBride 2017). This indicates that syntactic comprehension and vocabulary are at least partially separate constructs. However, a limitation to these studies is that the analyses were based on observed variables, leading to attenuation of correlations between variables (Nunnally and Bernstein 1997). In hierarchical regression analyses, this may lead both to underestimating the unique contribution of syntactic comprehension (due to error in the syntactic comprehension measure) and to overestimating the unique contribution of syntactic comprehension (due to error in the control variables).

1.2 | The Dimensionality of Syntax and Vocabulary

The question of whether syntactic comprehension plays a unique role in reading comprehension beyond vocabulary depends in part on the dimensionality of syntactic comprehension and vocabulary; that is, whether syntactic comprehension represents a unique source of individual differences in language comprehension—distinct or partially distinct from vocabulary. In other words, do we find children with relatively poor syntactic comprehension but good vocabulary and vice versa? On the one hand, there is substantial theoretical and empirical support for vocabulary and syntax being closely related (Croft 2000; Langacker 2016; Tomasello 2003), and it may seem reasonable that children develop their syntax and vocabulary under the influence of the same environmental sources. On the other hand, there are also reasons to expect that syntactic comprehension and vocabulary may vary partially independently from each other across individuals. For example, vocabulary could be more dependent on environmental factors than the development of syntactic comprehension. Languages contain a million words or more, but there is a limited set of rules or syntactic constructions (Chomsky 2002; Croft and Cruse 2004). Studies have shown that vocabulary is strongly related to socioeconomic status, in part due to large differences in the number of different words children are exposed to both through caregiver speech and book reading (Fernald et al. 2013; Hart and Risley 1995; Logan et al. 2019). In this way, syntactic comprehension may be less affected by the environment than vocabulary simply because there may be less variance in the syntactic structures children are exposed to. This notion is supported by several twin studies that found that although vocabulary and syntax were affected by the same genetic influences, vocabulary was less dependent on genes and more dependent on shared environment than syntax (Dale et al. 2000; Dionne et al. 2003; Mimeau et al. 2018). A second possible mechanism behind a partial distinction between vocabulary and syntactic comprehension is that the processing of vocabulary and syntax may be based on separate neurological processes: declarative and procedural memory, respectively (Ullman 2004). The present study was designed to reveal to which extent syntactic comprehension and vocabulary vary independently from each other across individuals, but cannot inform on the mechanisms behind such dimensionality of syntactic comprehension and vocabulary.

A growing number of studies have examined the dimensionality of lower language skills, especially the dimensionality of grammar more generally (without clear distinction between

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morphology and syntax) and vocabulary (Anthony et al. 2014; Language and Reading Research Consortium 2017; Language and Reading Research Consortium, Jiang, et al. 2018; Language and Reading Research Consortium, Yeomans-Maldonado, et al. 2018; Foorman et al. 2020; Language and Reading Research Consortium 2015; Lonigan and Milburn 2017; Tomblin and Zhang 2006; Wilson and Bishop 2021; Zhang 2012). The majority of studies found evidence that grammar and vocabulary either represent a single construct (Anthony et al. 2014; Language and Reading Research Consortium, Jiang, et al. 2018; Language and Reading Research Consortium, Yeomans-Maldonado, et al. 2018) or separate constructs that were so highly correlated that there was only limited support for discrimination between them (Lonigan and Milburn 2017; Tomblin and Zhang 2006). Although the Language and Reading Research Consortium (2015) and Tomblin and Zhang (2006) found evidence that grammar and vocabulary become more separate with age, the correlations remained very high even in the oldest age groups (*r*'s at 0.78 and 0.90). However, grammar is most commonly operationalized as a combination of syntactic and morphological knowledge, and therefore grammar variables often encompass potentially quite different skills, including both sentence-level (syntax) and word-level (morphology) measures. Measures of morphological knowledge typically target word-level knowledge such as the correct inflection of verbs and nouns, while syntax measures target sentence-level processes such as the comprehension or production of specific syntactic structures. When measures and latent constructs of grammar in this way have included word-level knowledge and processes, it is perhaps not surprising that it has been difficult to separate grammar from vocabulary in studies of dimensionality.

To our knowledge, only three studies have examined the dimensionality of syntax and vocabulary more specifically using latent variable modeling: Foorman, Herrera, et al. (2015), Foorman, Koon, et al. (2015), and Kieffer et al. (2016). These studies examined the dimensionality of syntactic comprehension and vocabulary in relation to reading comprehension and various other constructs, including phonological awareness, decoding skill, listening comprehension, and morphological awareness. The three studies reached the same overall conclusion: Syntax and vocabulary largely reflected a single construct, and syntactic comprehension did not explain unique variance in reading comprehension after controls. More specifically, Foorman, Herrera, et al. (2015) found that vocabulary and syntax represented a single construct either as a first- or second-order factor in kindergarten through second grade. Foorman, Koon, et al. (2015) and Kieffer et al. (2016) examined fourth to tenth grade and third to fifth grade, respectively, and found that the dimensionality of the data was best represented by bifactor models, consisting of both a general language factor that reflected syntactic comprehension and vocabulary and additional separate factors of syntactic comprehension and vocabulary. These results indicate that although there is some unique shared variance between syntax measures, there is a substantial overlap between syntax and vocabulary, providing little support for viewing syntactic comprehension and vocabulary as separate constructs. For example, although Foorman, Koon, et al. (2015) fit a model with a latent syntactic comprehension variable, the syntactic comprehension variable did not load significantly onto either syntactic comprehension test.

However, the studies above used tests of syntactic comprehension that arguably invoke other linguistic and nonlinguistic skills in addition to syntactic comprehension, potentially confounding the latent syntax variable and overestimating the relationship between syntax and vocabulary. Both Foorman, Koon, et al. (2015) and Foorman, Herrera, et al. (2015) used the Recalling Sentences subtest from the CELF-4 (Semel et al. 2003). Previous studies have found that although sentence repetition tasks do tap into syntactic comprehension, they also tap into vocabulary knowledge (Leclercq et al. 2014). In addition to Recalling Sentences, Foorman, Herrera, et al. (2015) used the Sentence Structure subtest from the CELF-4 (Semel et al. 2003), and Foorman, Koon, et al. (2015) used a grammatical judgment test. The Sentence Structure subtest uses a multiple-choice picture-selection format that is substantially dependent on nonlinguistic complex reasoning skills (De Cat and Melia 2022; Frizelle et al. 2017; Authors 2024). Grammatical judgment tests arguably tap into syntactic awareness rather than syntactic comprehension (Brimo et al. 2018). Finally, Kieffer et al. (2016) used a single measure of syntactic comprehension, namely the Formulated Sentences subtest from the CELF-4 (Semel et al. 2003). In this test, participants have to produce grammatically, semantically, and logically acceptable sentences that include specific target words and describe specific pictures. At face value, this task involves more than just syntax. The participant also has to attend to the pictures, understand the target words, and generate sentences. In fact, the scoring instructions put more emphasis on producing a sentence that accurately describes the picture than on producing syntactically correct sentences: Sentences that do not fit the pictures are given zero points, while sentences that contain up to two grammatical errors are given one point (Semel et al. 2003).

The advantage of latent variable modeling is that it allows for more precise estimation of constructs by modeling the shared variance between two or more indicators. This partitions out random error variance and variance caused by the specific test format from variance due to the target construct (Brown 2015). However, this only works to the extent that the shared variance between indicators represents the target construct (Rhemtulla et al. 2020). In the three studies above, the syntactic comprehension tasks likely involved additional skills related to syntactic comprehension, including vocabulary and fluid-reasoning skills, making it difficult to determine whether the shared variance between syntax measures actually reflects syntactic comprehension. Thus, it is unclear if the models accurately represent the relationship between syntactic comprehension, vocabulary, and reading comprehension.

1.3 | Present Study

The goal of the present study was twofold. First, we investigated whether syntactic comprehension and vocabulary constitute separate sources of individual differences in language comprehension by examining the dimensionality of syntactic comprehension and vocabulary measures. Second, we examined if individual differences in syntactic comprehension were related to reading comprehension by examining if syntactic comprehension explained variance in reading comprehension after controlling for vocabulary and decoding skill. A special feature

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of the design was that syntactic comprehension and vocabulary each were assessed with both expressive and receptive measures to balance method variance having to do with the modality of input. Syntactic comprehension was additionally assessed with a third measure in the written modality.

The study used a sample of Danish sixth-grade children because, up until this age, reading comprehension is largely dependent on decoding skill (Florit and Cain 2011), especially considering that Danish is a relatively opaque orthography (Seymour et al. 2003). At this age, there is also evidence that comprehension subcomponents have started to separate out into different dimensions (Language and Reading Research Consortium, Jiang, et al. 2018; Language and Reading Research Consortium, Yeomans-Maldonado, et al. 2018; Tomblin and Zhang 2006) and school texts become increasingly complex, which may put more pressure on the ability to use syntactic signals to understand texts. This study handles measurement error both statistically by using latent variable modeling and by design by using syntax tests specifically designed to measure syntactic comprehension.

2 | Method

2.1 | Participants

In total, 190 Grade 6 students from three public schools in the Greater Copenhagen area, Denmark, participated in the study in the fall of 2021. This was during the COVID-19 pandemic, but schools were open at this point. Danish is the majority language of Denmark. Twenty-nine students were bilingual and were not included in the analyses in order to avoid any confounds from second-language learning. The remaining 161 students (77 girls and 84 boys) had a mean age of 12;3 (SD = 0;4). We intended a sample size of 200 majority language speakers, but fewer students wanted to participate than anticipated, probably due to the COVID situation. The students were tested twice individually by trained student assistants in a quiet room at their school during school hours. The reading comprehension tests were administered in groups. All testing took place in person. Only children with written consent from a parent participated in the study, and students gave their assent prior to testing. The study was approved by the Research Ethics Committee at the University of Copenhagen in accordance with relevant guidelines and legislation, including the Declaration of Helsinki and The European Code of Conduct for Research Integrity.

2.2 | Measures

2.2.1 | Syntactic Comprehension

Syntactic comprehension was measured with three researcher-created tests: an expressive sentence repetition test, a receptive written whodunnit test, and a receptive auditory whodunnit test. The items in all three tests were designed to be semantically reversible, so children had to rely on syntactic signals rather than background knowledge. Although sentence repetition is an expressive test, studies have found that solving sentence repetition tasks builds on sentence comprehension (Lombardi and Potter 1992; Potter and Lombardi 1998; Riches 2012), although less than the whodunnit task (Authors 2024). As sentence repetition tests have previously been shown to also tap into vocabulary knowledge (Leclercq et al. 2014), extra care was taken in the design of the sentence repetition test to minimize vocabulary demands and increase demands on syntactic comprehension. This was done by using simple vocabulary and including difficult syntactic constructions such as passives, object-extracted relative clauses, and object-verb-subject constructions, which are known to cause comprehension difficulties (Boeg Thomsen and Poulsen 2015; Dabrowska and Street 2006; Ferreira 2003; Kristensen et al. 2014). The two whodunnit tests were also designed with simple vocabulary and difficult syntactic constructions. All three syntactic comprehension tests were administered on a laptop computer using OpenSesame experiment software version 3.3 (Mathot et al. 2012). Items were presented in a fixed quasi-randomized order. There were no discontinuation rules in any of the three syntactic comprehension tests.

In the sentence repetition test, the students heard pre-recorded sentences, which they were asked to repeat verbatim. There were thirty sentences representing six syntactic constructions. The included syntactic constructions and examples of items are presented in Table 1. The score was the proportion of correctly repeated sentences.

The auditory whodunnit test was identical to the sentence comprehension test used in Authors (2022). The students first heard a pre-recorded sentence (e.g., "George was kicked by Sam under the table"), and then heard a question regarding the syntactically coded information in the sentence, for example "who kicked?" They responded by pressing one of two keys corresponding to two written response options shown on the screen (here "George" and "Sam"). The questions could not be correctly answered based on background knowledge and thus required the student to establish meaning relations between words solely based on syntactic signals. The test contained 60 items representing five syntactic constructions, and the score was the proportion of correct answers. The five syntactic constructions are presented in Table 2 with examples of items.

The written whodunnit test was nearly identical to the auditory sentence comprehension test, with the exception that the sentences and questions were presented in writing on the screen rather than auditorily. The items were different but represented the same five syntactic constructions. Because the two whodunnit tests were similar in format, we expected to see a learning effect from one test to the other. For this reason, the order of administration was balanced across students, with half of the students given the written test first and the other half given the auditory test first.

2.2.2 | Vocabulary

Vocabulary was measured with a researcher-created expressive vocabulary test (Gellert and Vang Christensen 2012) and a Danish adaptation of the receptive Peabody Picture Vocabulary Test (PPVT) (Dunn 1959; Nielsen 2008). In the expressive vocabulary test, the student was presented with 90 pictures of objects and had to name the object in each picture. In this test, all students were presented with all items. In the adapted PPVT,

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TABLE 1 | Syntactic constructions in the sentence repetition test.

Construction	#Items	Example sentences and translations
Active SVO	4	"Det underlige væsen opdagede pigen på slottet" (The weird creature discovered the princess in the castle)
Passive	4	"Sygeplejersken med det lange hår blev omfavnet af den blege patient" (The nurse with the long hair was embraced by the pale patient)
Active OVS	6	"Dronningen ville kongen nok tilbyde den varme suppe" (The king would probably offer the queen the warm soup) (word-for-word: "The queen would the king probably offer the warm soup")
Subject-extracted relative clause	4	"I skyggen hvilede bedstemoderen som havde kildet barnet" (The grandmother that had tickled the child rested in the shade) (word-for-word: "In the shade rested the grandmother that had tickled the child")
Object-extracted relative clause	8	"På skibet legede katten som rotten havde bidt" (The cat that the rat had bitten played on the ship) (word-for-word: "On the ship played the cat that the rat had bitten")
Oblique-extracted relative clause	4	"Det gule bord som vasen med blomster står på skal males" (The yellow table that the vase with flowers is standing on needs to be painted)

Note: English translation in parentheses. Word-for-word translations are added when Danish word order differs from English.

the students were asked to point to one of four pictures that best represented a word spoken by the examiner. There were 150 items, which increased in difficulty, and the test was discontinued after six incorrect responses across eight consecutive items. In both vocabulary tests, the score was the proportion of correct items out of all possible items.

2.2.3 | Decoding Efficiency

Decoding efficiency was measured with a word and a non-word list from *Elbros ordlister* ("Elbro's Word Lists") (Elbro 1990). The lists consisted of twenty items each, and the score was the number of correct responses divided by the time it took to read the list. We used this efficiency score rather than decoding accuracy because we expected a strong ceiling effect on the accuracy score in this age group. This turned out to be the case as the average accuracy was 96% and 88% for the two lists in the present sample. The data included a third decoding measure from the EVALD test material (Petersen and Rønberg 2020), but based on preliminary analyses, the format of this measure was too dependent on language comprehension in the current age group for the purposes of the present study[1] and was not included in the main analyses.

2.2.4 | Reading Comprehension

Reading comprehension was measured with two tests: A standardized test, *Tekstlæseprøve 8* ("Text reading test 8") (Møller 2015) and a Danish translation of part of the Diagnostic Reading Analysis (DRA) test (Crumpler and McCarty 2004; Daugaard et al. 2017). Text Reading Test 8 (TRT 8) is a connected expository text on a single topic with 11 yes-no comprehension questions and 43 cloze items with four options each. The children were given 35 min to complete the test. Seventy-two percent of students completed all items within the time limit. The adapted DRA test consisted of eight short expository and narrative texts with between five and six open-ended comprehension questions for each text. There were a total of 48 questions, and the students were given 30 min to complete as many as possible. Twenty-seven percent of students completed all items within the time limit. For both reading comprehension tests, the score was the proportion of correct items out of all possible items.

2.3 | Analyses

The dimensionality of syntactic comprehension and vocabulary and the relationship between syntactic comprehension and reading comprehension were analyzed using Confirmatory Factor Analysis and Structural Equation Modeling. All analyses were performed in R statistical software version 4.1.2 (R Core Team 2021) using the lavaan package version 0.6–0.17 (Rosseel 2012). We used maximum likelihood estimation with robust (Huber-White) standard errors and a scaled chi-squared (χ^2) test statistic to adjust for the slight nonnormality of the data. Missing data were handled using full information maximum likelihood. Model fit was assessed based on the χ^2 statistic, comparative fit index (CFI), Tucker–Lewis index (TLI), root mean square error of approximation (RMSEA), and standardized root mean squared residual (SRMR). A non-significant χ^2 statistic, CFI $\geq$ 0.95, TLI $\geq$ 0.95, RMSEA $\leq$ 0.06 and an SRMR $\leq$ 0.08 were interpreted as indicative of excellent fit (Brown 2015; Hu and Bentler 1999). Model comparison was performed using the scaled chi-square difference test and by comparing Akaike information criterion, where smaller values indicate better fit. In addition to factor analyses, the dimensionality of syntactic comprehension was assessed by calculating the average variance extracted (AVE) (the average amount of variance a factor explains in its indicators) and construct reliability (CR) (the total amount of variance in all indicators explained by the factor). AVE values above 0.50 and CR values above 0.70 were considered indicative

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TABLE 2 | Syntactic constructions in the auditory whodunnit test.

Construction	#Items	Example sentences and translations	Question	Options
Active SVO	20	"Rasmus sparker Christian ved bordet" (Rasmus kicks Christian at the table)	"Hvem sparker?" (Who kicks?)	"Rasmus" "Christian"
Passive	10	"Spionen har sekretæren allerede takket" (The secretary has already thanked the spy) (word-for-word: "The spy has the secretary already thanked")	"Hvem takkede?" (Who thanked?)	**"Sekretæren"** (The secretary) "Spionen" (The spy)
Premodified noun	10	"Den store kanin jagtede grisen på marken" (The big bunny chased the pig on the field)	"Hvilken en var. stor?" (Which one was big?)	**"Kaninen"** (The bunny) *"Grisen"* (The pig)
Embedded relative clause	10	"Spanden som posen ligger i om natten er sort" (The bucket that the bag lies in at night is black)	"Hvilken en var. sort?" (Which one was black?)	**"Spanden"** (The bucket) "Posen" (The bag)
OVS	10	"Soldaten har politimanden nok hjulpet" (The police officer has probably helped the soldier) (Word-for-word: "The soldier has the police officer probably helped")	Hvem hjalp? (Who helped?)	**"Politimanden"** (The police officer) "Soldaten" (The soldier)

Note: English translations in parentheses. Word-for-word translations are added when Danish word order differs from English. Correct answers are marked in bold.

of convergent validity of a factor (Fornell and Larcker 1981; Hair et al. 2010). Discriminant validity was assessed using the Fornell-Larcker criterion, which states that a factor exhibits discriminant validity when its AVE is higher than the squared correlation with other factors; in other words, when the factor explains more variance in its indicators than it shares with other factors (Fornell and Larcker 1981). We tested for a unique association between reading comprehension and syntactic comprehension by fitting a structural equation model where reading comprehension was set to regress onto decoding, vocabulary, and syntactic comprehension. A significant regression coefficient from reading comprehension to syntactic comprehension was considered indicative of a unique association between syntactic comprehension and reading comprehension. In Section 3, we outline the specific steps involved in model building. Given that the planned analyses yielded satisfactory model fits, no post hoc adjustments were made except for the exclusion of the decoding measure described above in Section 2.2.3.

2.4 | Transparency and Openness

We report the methods of sample selection, data exclusions and manipulations, and all measures in the study. Data were analyzed using R, 4.1.2 (R Core Team 2021) and the lavaan package version 0.6-19 (Rosseel 2012). This study was not preregistered, but deviations from planned analyses are described. Analysis code and the final dataset are available at https://osf.io/vux7f/?view_only=bafc71d6409f4e92bad97588b02a4f36.

3 | Results

Sixteen students were missing the reading comprehension assessments because they were not present at school on the days when tests were administered. The reading comprehension tests were the last to be administered, and a rise in COVID infections limited data collection. A further two students misunderstood the instructions to the TRT 8, and their results were discarded. In total, 18 students (11.2%) were missing the TRT 8, and 16 students (9.9%) were missing the DRA. Six students were missing one of the individually administered tests. In total, 84% of students completed all tasks. One student was a multivariate outlier on the vocabulary tests, and had these scores removed. Another student had a univariate outlier value in expressive vocabulary ($z < -3.39$), and had the score replaced with the next lowest score. In total, 3.3% of data was missing. Based on Little's missing completely at random (MCAR) test (Little 1988), the data were MCAR ($\chi^2(54) = 56.92$, $p = 0.367$). We estimated models using full information maximum likelihood, allowing us to use all available data without excluding students who were missing one or more tests.

Table 3 shows the descriptive statistics for each measure. All measures had skew and kurtosis values within ±1. For all measures, the reported reliability is calculated based on the present sample. In the receptive vocabulary and reading comprehension tests, not all students completed all items due to a discontinuation rule and time limits, respectively. For this reason, the reliability could not be calculated based on all items. In these three tests, the reliability was calculated based on the set of items that

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at least 80% of students had completed before reaching the discontinuation rule or timing out.

As shown in Table 4, the correlations between syntax measures were high (0.55–0.61), while the correlation between vocabulary measures was only moderate (0.39). The correlations between syntax measures and vocabulary measures ranged from weak (0.16) to moderate (0.39). The two decoding measures only correlated weakly with the vocabulary measures, which are not uncommon in Scandinavian studies (Daugaard et al. 2017; Gellert et al. 2020; Hjetland et al. 2019; Lervåg et al. 2018; Poulsen and Gravgaard 2016; Authors 2022).

To formally assess the dimensionality of syntactic comprehension and vocabulary, we estimated two confirmatory factor models. In Model A, the syntax and vocabulary measures loaded onto one factor, and in Model B, the syntax and the vocabulary measures loaded onto separate factors. The model fit statistics for both models are shown in Table 5. Because the goal was to examine the dimensionality of syntactic comprehension and

TABLE 3 | Descriptive statistics for all measures.

Measures	*n*	M	SD	Range	Skew	Reliability
Sentence repetition (prop.correct)	161	0.45	0.20	0–0.97	0.03	0.87[a]
Aud. whodunnit (prop.correct)	160	0.79	0.12	0.47–0.98	−0.46	0.83[a]
Writ. whodunnit (prop.correct)	160	0.82	0.10	0.5–1	−0.62	0.82[a]
Expressive vocabulary (prop.correct)	160	0.74	0.08	0.5–0.9	−0.63	0.85[a]
Receptive vocabulary (prop.correct)	160	0.41	0.11	0.17–0.74	0.67	0.82[b]
Word decoding (correct/sek.)	160	1.39	0.42	0.34–2.41	0.15	0.85[c]
Nonword decoding (correct/sek.)	160	0.78	0.27	0.06–1.58	0.20	0.85[c]
TRT8 (prop. correct)	143	0.61	0.16	0.26–0.94	−0.21	0.80[d]
DRA (prop.correct)	145	0.57	0.15	0.2–0.93	−0.06	0.78[e]

Note: TRT 8 = text reading test 8, DRA = diagnostic reading analysis.
[a]Cronbach's alpha of all items.
[b]Cronbach's alpha based on the first 50 items, which 83% of students completed.
[c]Spearman Brown-corrected correlation between Word List and Nonword List.
[d]Cronbach's alpha based on the first 48 items, which 80% of students completed.
[e]Cronbach's alpha based on the first 32 items, which 80% of students completed.

TABLE 4 | Zero-order correlations between all measures.

	1	2	3	4	5	6	7	8
1. Sentence repetition	—							
2. Aud. whodunnit	0.59	—						
3. Writ. whodunnit	0.55	0.61	—					
4. Expressive vocabulary	0.33	0.27	0.26	—				
5. Receptive vocabulary	0.33	0.23	0.16	0.39	—			
6. Word decoding	0.29	0.29	0.25	0.02	0.00	—		
7. Nonword decoding	0.41	0.36	0.27	0.06	0.10	0.71	—	
8. TRT 8	0.55	0.60	0.53	0.40	0.29	0.35	0.40	—
9. DRA	0.50	0.54	0.48	0.28	0.10	0.38	0.44	0.65

Note: Listwise deletion. Correlations ≥ 0.17 are significant at $p < 0.05$.

TABLE 5 | Fit statistics for one- and two-factor models.

Model	χ^2	Scaling factor	df	*p*	RMSEA [90% CI]	SRMR	CFI	TLI	AIC
A: One-factor	16.942	1.192	5	0.005	0.135 [0.071–0.206]	0.061	0.925	0.850	−1353.96
B: Two-factor	6.198	1.221	4	0.185	0.066 [0.000–0.158]	0.034	0.986	0.964	−1364.59

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vocabulary, only the syntactic comprehension and vocabulary measures were included in these models. This way, we ensured that the fit statistics of the two models specifically reflect the relationship between vocabulary and syntax and were not affected by the inclusion of other variables.

The fit of Model A provided a poor to unsatisfactory fit, while Model B had an excellent fit. Because the two models can be considered nested, we compared them using the scaled chi-squared difference test (Brown 2015; Kline 2016). Based on this test, Model B fit the data significantly better than Model A (scaled $\Delta\chi^2 = 11.73$, $df = 1$, $p < 0.001$), and AIC was lower for Model B. Model B is shown in Figure 1 with standardized parameter estimates.

Inspection of the local fit of model B revealed a correlated residuals value above 0.10 for sentence repetition and receptive vocabulary test. Adjusting Model B by allowing the residuals of these two indicators to be correlated ($r = 0.19$) improved the fit significantly (scaled $\Delta\chi^2 = 4.51$, $df = 1$, $p = 0.03$). But all factor loadings were within 0.05 of the original model, which we retain for now because we have no theoretical reasons to allow the correlated residuals.

The AVE and CR of the syntax factor were 0.55 and 0.78, respectively, indicating convergent validity. The syntax factor also exhibited discriminant validity by satisfying the Fornell-Larcker criterion with an AVE (0.55) larger than the squared correlation with vocabulary (0.31). The AVE and CR of the vocabulary factor were low at 0.35 and 0.52, respectively. However, vocabulary also exhibited discriminant validity based on the Fornell-Larcker criterion. Thus, syntactic comprehension and vocabulary seem to represent separate, but correlated, constructs.

In order to examine the relationship between syntactic comprehension and reading comprehension, we first fit a measurement model with four latent variables of vocabulary, syntactic comprehension, decoding, and reading comprehension to assess whether the measures reflected the expected factor structure. This model provided an excellent fit to the data (scaled $\chi^2 = 23.63$ ($df = 21$, $p = 0.31$, scaling factor = 1.02), CFI = 0.997, TLI = 0.995, RMSEA = 0.022 [0.000, 0.075], SRMR = 0.036). In the next step, we set reading comprehension to regress onto decoding skill, vocabulary, and syntactic comprehension. This model had identical fit statistics to the measurement model. Inspection of local fit again revealed a correlated residual value above 0.10 between receptive vocabulary and sentence repetition. Adjusting the model by allowing the residuals to correlate ($r = 0.20$) significantly improved model fit (scaled $\Delta\chi^2 = 6.42$, $df = 1$, $p = 0.01$). However, this had little impact on the estimates of factor loadings or direct effects on reading comprehension (differences less than 0.03). Since there is no theoretical justification for allowing the correlations, and the adjustment has little impact on the results, we report the planned, unadjusted structural model in Figure 2 with standardized parameter estimates. All three predictors explained unique variance in reading comprehension, but syntactic comprehension was the strongest predictor by far ($\beta = 0.63$ compared to $\beta = 0.24$ and $\beta = 0.23$ for vocabulary and decoding). The model explained 80% of the variance in reading comprehension in total.

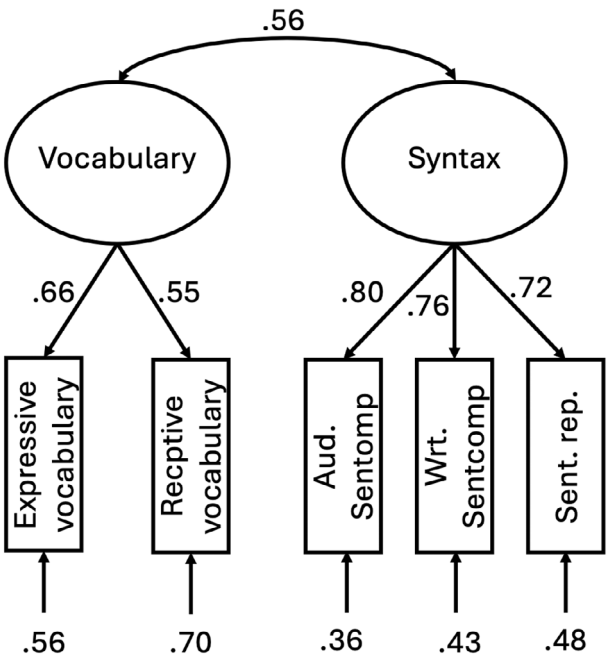


FIGURE 1 | Model B: Two-factor model of syntactic comprehension and vocabulary. All parameter estimates are significant at $p < 0.05$.

4 | Discussion

The present study found that syntactic comprehension and vocabulary represented separate, correlated constructs and that syntactic comprehension explained unique variance in reading comprehension beyond decoding skill and vocabulary. The fact that syntactic comprehension and vocabulary represented separate factors indicates that they constitute separate sources of individual differences in language comprehension. This means that syntactic comprehension skill and vocabulary size vary partially independently of each other across individuals. This result differs from some previous latent-variable studies of syntactic comprehension and vocabulary that found little evidence of separate syntactic comprehension and vocabulary constructs (Foorman, Herrera, et al. 2015; Foorman, Koon, et al. 2015; Kieffer et al. 2016). The results are more in line with studies finding separable but highly correlated constructs (Lonigan and Milburn 2017), but the present study found more moderate correlations between the constructs ($r = 0.56$). A substantial difference between the present study and previous studies is that the present study uses researcher-created syntactic comprehension tests specifically designed to isolate syntactic comprehension as much as possible, while the previous studies made use of standardized tests that likely tap into a wider range of linguistic and nonlinguistic skills as detailed in the beginning of the paper. This indicates that even when using latent-variable modeling, the use of tests not specifically designed to measure syntactic comprehension may blur distinctions between constructs, perhaps especially when examining the dimensionality of closely related constructs.

While the present data cannot inform on the mechanisms behind the distinction between syntactic comprehension and vocabulary, the emergence of syntax and vocabulary as separate factors suggests that they, to some extent, are acquired or processed in different ways. This is in line with previous twin studies that found that vocabulary development is more dependent on the environment than the development of syntax (Dale et al. 2000; Dionne et al. 2003; Mimeau et al. 2018). It is worth

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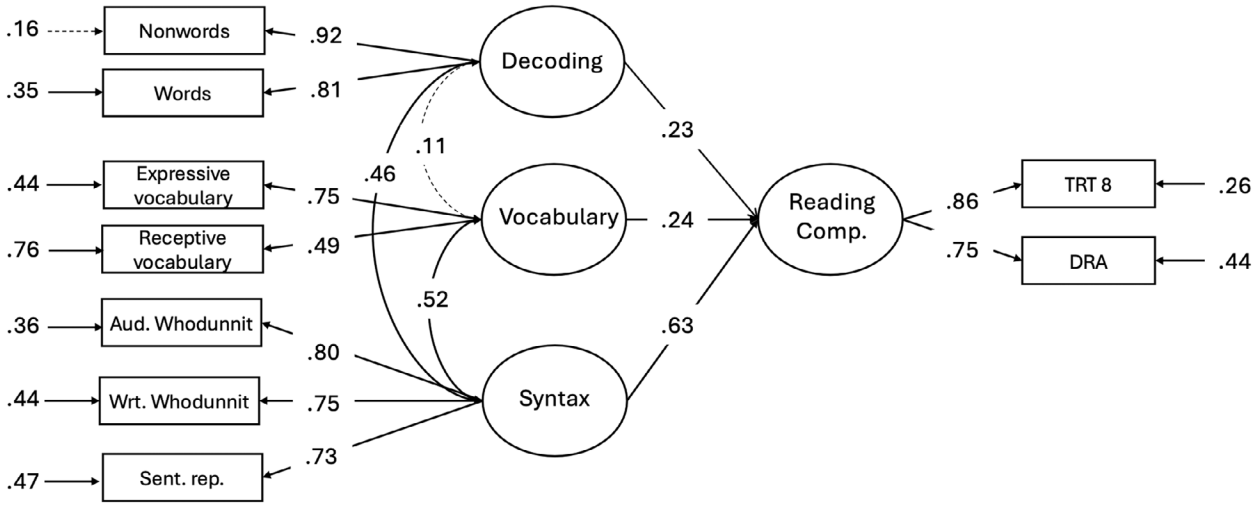


FIGURE 2 | Structural equation model of decoding, vocabulary, syntax, and reading comprehension. Solid lines represent parameter estimates that are significant at $p < 0.05$, while dotted lines represent nonsignificant estimates.

noting that there are only a few studies of the etiology of syntax and vocabulary among school-aged children, and the results of these studies are mixed. While DeThorne et al. (2012) and Dale et al. (2010) did not find evidence of any differences between syntax and vocabulary, Mimeau et al. (2018) reported a substantial difference between syntax and vocabulary where shared environment explained about half of the variance in vocabulary but no significant variance in syntax. Future studies might focus on further disentangling the etiology of vocabulary and syntactic comprehension in school-aged children, as the present study suggests that slightly different mechanisms of acquisition may become visible in this age group.

4.1 | The Relationship Between Syntactic Comprehension and Reading Comprehension

The present study extended previous studies' findings of separable constructs (Lonigan and Milburn 2017) by showing that syntactic comprehension explained unique variance in reading comprehension when controlling for decoding skill and vocabulary. This result is consistent with several previous studies (Poulsen and Gravgaard 2016; Authors 2022; Silva and Cain 2015; Sorenson Duncan et al. 2021; Tong and McBride 2017). The present study is correlational and cannot determine if there is a causal relationship between syntactic comprehension and reading comprehension. However, the fact that syntactic comprehension explained unique variance in reading comprehension suggests that some children may experience difficulties comprehending text not because they do not know the individual words, but because they have difficulties using syntactic signals to derive the syntactic meaning relations between individual words.

In the present study, syntactic comprehension was by far the best predictor of reading comprehension ($\beta = 0.63$ compared to $\beta = 0.24$ and $\beta = 0.23$ for vocabulary and decoding, respectively). Considering the strong empirical support for the importance of vocabulary to reading comprehension (National Reading Panel 2000; Quinn et al. 2015; Spencer et al. 2017), it is surprising that syntactic comprehension was such a strong predictor compared to vocabulary, especially given the fact that syntactic comprehension has received relatively little attention in literacy research so far. However, this result is not unprecedented. Both Poulsen and Gravgaard (2016), Sorenson Duncan et al. (2021), and Silva and Cain (2015) report higher standardized final beta coefficients for syntax than for vocabulary in hierarchical regression analyses predicting reading comprehension. One possible interpretation of this result is that the biggest obstacle for the majority of typically developing children in this age group is not understanding individual words in texts, but using syntactic signals to connect the word meaning into sentence meaning. The majority of children will probably have little difficulty with most of the words they encounter during reading, but syntactic signals guide literal comprehension in every single sentence. Vocabulary, background knowledge, and inferencing will also help integrate word meanings into a complex comprehension model (Perfetti and Stafura 2014), but this route to comprehension depends on predictability and redundancy. The point of texts and reading is also to experience and learn new things, and in this process, attention to syntactic signals may help establish literal meaning. It is also possible that robust syntactic comprehension partially compensates for weaker vocabulary. Children who are good at using syntactic signals may depend less on guessing the meaning of sentences based on the knowledge of individual words, and therefore their reading comprehension may be less vulnerable to encountering words they do not know. For example, if children have robust syntactic comprehension, they are likely to understand who is doing what to whom in a sentence such as "Albert was eaten by the stynx with very sharp teeth" without knowing exactly was a stynx is. Finally, it is possible that different reading comprehension tests rely on vocabulary to different extents depending on the difficulty of the vocabulary in the texts. While one of the reading comprehension tests in the present study, the DRA, consisted of texts with mainly simple vocabulary, the other reading comprehension test, the TRT 8, was an expository text that contained difficult, subject-specific vocabulary. In this way, the latent reading comprehension variable represents a combination of low and high vocabulary texts that are considered appropriate

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for the age group. As there are no standards for how vocabulary dependent a reading comprehension test should be, it is difficult to determine whether the contribution from vocabulary is over- or underestimated. The present study cannot disentangle how syntactic comprehension and vocabulary interact, but the results do indicate that individual differences in syntactic comprehension are at least as important to reading comprehension as individual differences in vocabulary. Future research needs to evaluate how syntactic comprehension and vocabulary interact in relation to reading comprehension.

4.2 | Limitations

Arguably, the sentence-level measures could depend on working memory resources, and as there were no working memory control measures, the results do not rule out that the separation between syntax and vocabulary is underpinned by working memory. But previous studies have often found that the correlations of non-linguistic working memory measures and both sentence comprehension and reading comprehension are weak to moderate (Oakhill and Cain 2012; Poulsen and Gravgaard 2016; Sorenson Duncan et al. 2021), and thus it is unlikely that working memory is responsible for substantial amounts of the variance in syntactic comprehension that is relevant for reading comprehension (cf. Authors 2022, for discussion). For this reason, working memory measures were left out of the present study to reduce test fatigue. However, future studies including additional latent variables of non-linguistic cognitive skills would strengthen this interpretation.

Another limitation is that our two decoding tests used the same format (list reading), meaning that we did not control for method variance in the latent decoding variable. As mentioned, we had originally planned to use a third decoding measure in a different test format. However, it was left out of the present analyses because preliminary factor analyses revealed that this test was difficult to fit into the models as an indicator of decoding because it similarly correlated with several of the other measures. The omitted measure is included in the available data set (see Section 2.2.6).

Finally, the correlation between vocabulary measures was low ($r = 0.36$). Forming latent variables on the basis of poorly correlated tests may inflate the relations with other latent variables if the shared variance between the indicators does not represent the target construct (Rhemtulla et al. 2020). There is, however, empirical evidence that expressive and receptive vocabulary represent the same construct in this age group (Lonigan and Milburn 2017; Tomblin and Zhang 2006). Furthermore, even if the correlation between vocabulary and syntactic comprehension were inflated, the fact that syntax and vocabulary emerge as separate constructs despite this inflation supports the conclusion that vocabulary and syntactic comprehension represent partially distinct constructs.

5 | Conclusion

The present study found evidence that syntactic comprehension and vocabulary constitute separate sources of individual differences in language comprehension. In comparison with previous latent-variable studies, this result suggests that it is easier to tease apart slightly different dimensions of language comprehension when employing tests specifically designed to measure those dimensions. The study also confirmed the mounting evidence that individual differences in syntactic comprehension are strongly correlated with a reading comprehension. This suggests that some children misunderstand texts because they have difficulty establishing how individual words relate to each other in sentences based on syntactic signals rather than because they do not know the word meanings. Indeed, the results suggest that individual differences in syntactic comprehension are at least as important to reading comprehension as individual differences in vocabulary, indicating that syntactic comprehension deserves substantially more attention in literacy research and educational practice than it has previously received. More research is needed to establish causality, but the opportunity for intervention is interesting: It is difficult to find principles of leverage in background knowledge or vocabulary learning and teaching; each word is for a large part a new subject. Syntax is for a large part principled—a *limited* set of rules or templates govern how word order signals propositional meaning. This raises the possibility of leverage: That limited teaching and learning will be applicable and useful in many cases. The challenge is to find the right targets to focus on or to discover how to make children more reliably use the syntactic knowledge they already have.

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Ethics Statement

The study was approved by the Research Ethics Committee at the University of Copenhagen [journal number: 504-0151/24-4000] in accordance with relevant guidelines and legislation.

Conflicts of Interest

The authors declare no conflicts of interest. An earlier version of this article was included as part of the first author's PhD dissertation.

Data Availability Statement

The data and analysis code are available at https://osf.io/vux7f/?view_only=bafc71d6409f4e92bad97588b02a4f36.

Endnotes

[1] The EVALD decoding test was in an odd-one-out format where children were presented with five different written words and had to choose the word that did not fit in semantically, for example, "house" in "black," "blue," "house," "green," "yellow." There were 87 items, and the children were given 4 min to complete as many items as possible. Although this task arguably also draws on semantic knowledge, the test is constructed with very easy vocabulary items with the intention for it to be a decoding test. However, a latent measurement model with the EVALD test as an indicator of decoding skill did not provide an acceptable fit to the data. An inspection of the modification indices and the standardized residuals of the model revealed that the poor model fit was due to residual correlations between the EVALD test and the many of the other measures, and indeed the zero-order correlations with the syntactic measures were similar to the correlations

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with decoding. As the decoding construct is of lesser importance for the present purposes, and because it is unclear what the test measures and how to fit it into the models, it was left out of the main analyses. But the data are available in the provided data set.

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